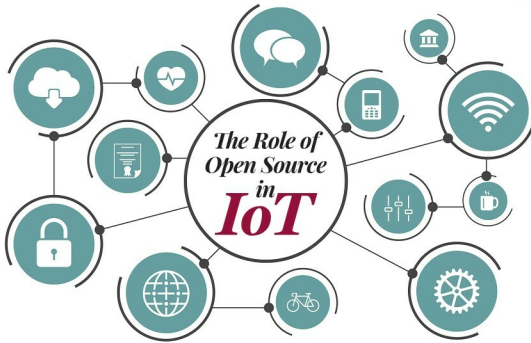




IoT allows different objects to be sensed or controlled remotely across the existing network infrastructure. It also helps to integrate objects from the physical world with computer-based systems and, hence, results in improved accuracy, efficiency and economic benefits, as well as reduced human intervention in their operation.

Hurrying for a party, we may forget to switch off the fans and lights at home. And remember that lapse only half way through the party. Returning home to switch off the fans and lights is simply not an option. Wouldn't it be useful if we could switch off the gadgets and lights at home even when away? What if we could switch on our air conditioners even before entering our homes so that we don't have to wait for the room to become cooler? Around 20 years ago, this would have just been a pipe dream but, today, this is a reality because of IoT or the Internet of Things, which amalgamates software with different physical hardware devices.

Wikipedia defines IoT as the internetworking of physical devices (also called smart devices and connected devices), vehicles, buildings and anything else embedded with electronics, sensors, actuators, software and network connectivity, which further enables these objects to collect and exchange data. In simple words, the Internet of Things refers to the vast world of different interconnected devices that have embedded sensors, which are capable of providing data and also being controlled through the Internet.

So now, even when not at home, we can switch off the fans and lights of our rooms using the Internet. Other common examples include different home automation devices, like remotely controllable lighting fixtures and thermostats. Then there are traffic sensors, water quality meters, smart electric grids and components that track manufactured goods, all of which work with the help of IoT.

The core concept of the Internet of Things became popular in 1999, through the Auto-ID Centre at MIT in the US. Since then, there has been rapid growth in the IoT space, with the emergence of a number of competing standards, projects, tools, frameworks, policies and organisations, which work towards defining how different connected devices communicate in the current era. Different open source tools and open standards are important and will continue to be so in the future, in order to ensure that all these devices are able to interconnect properly. These open source tools will also be responsible for the back-end tasks of processing the large volumes of Big Data that all of these devices will generate in the future.

IoT allows different objects to be sensed or controlled remotely across the existing network infrastructure. It also helps in creating different opportunities for objects in the physical world to directly integrate with computer-based systems, resulting in improved accuracy, efficiency and economic benefits as well as reduced human intervention. According to the estimates of several experts, IoT will consist of around 30 billion objects by 2020. It is expected to offer highly advanced connectivity to different devices, systems and services that will actually go beyond machine-to-machine (M2M) communication, covering a variety of domains, protocols and applications. The interconnection of such embedded devices is expected to usher in automation ubiquitously, and also enable various advanced applications like smart grids to expand to areas such as smart cities.